Implement Entra ID Integration with ABM for Account Auto Provisioning for Microsoft Intune — Government (GCC High)

Implementation Solution Guide

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Executive Summary — Entra ID + ABM Managed Apple ID Auto-Provisioning (Government / GCC High)

The bottom line

This is the **Government (GCC High)** edition of integrating Microsoft Entra ID with Apple Business Manager (ABM) for Apple device provisioning. The single most important fact for a decision-maker: **the Entra ■ ABM _federation_ and _SCIM auto-provisioning_ link does not support national clouds (GCC High / DoD) today.** Apple's documentation states, verbatim, "Integration with national clouds isn't currently supported," and Microsoft does not offer the ABM provisioning gallery app for the US Government cloud.

What this means: a pure GCC High tenant **cannot** deliver the "sign in to Apple devices with your Entra work account, auto-created and kept in sync by SCIM" experience the way a Commercial tenant can. That is a **feature parity gap**, not a portal-URL difference.

What is delivered instead — and is fully supported: Apple device management through **Intune Government** — ABM linked to Intune for **Automated Device Enrollment** and **User Enrollment**, with Managed Apple IDs created through ABM's own tooling (manual, SFTP, or ABM API). This achieves the device-management outcome government teams need, on a supported path, using the sovereign `*.us` portals and the US Gov Microsoft Graph endpoint.

Why this edition exists

Commercial implementation steps **silently fail** on GCC High: the admin URLs differ (`*.us`), the tooling must be told it is US Gov (`Connect-MgGraph -Environment USGov`), the licensing is Government SKUs, and — most importantly — the headline federation/SCIM capability is unavailable. A government administrator following a Commercial guide would waste effort configuring a flow that cannot complete. This edition sets expectations correctly and guides the supported path.

What you get

- A **decision gate** that tells you, before any spend, whether your requirement is achievable on GCC High today or depends on the unsupported federation/SCIM link.
- Ordered, novice-friendly **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** steps using the sovereign portals, with GCC-High caveats called out where they matter.
- **Five PowerShell scripts** (discovery, validation, reporting, safe remediation, scope targeting) that connect to `graph.microsoft.us` via `-Environment USGov`.
- A **cited parity reference**, **pass/fail testing gates**, and **GCC-High-specific troubleshooting**.

The risk it removes

Deploying with Commercial assumptions on a government tenant leads to failed enrollments, misconfigured identity, and — worst of all — stakeholders expecting a capability the sovereign cloud does

not offer. This edition delivers a **correct, supported, pilot-proven** configuration and a **documented parity gap**, so the outcome is right the first time and no one is surprised.

Sovereign-cloud capabilities change. The federation limitation is quoted from Apple's current documentation; re-verify the cited Apple and Microsoft pages (see `gcc-high-parity-notes.md`) before implementation.

Solution Overview — Entra ID + Apple Business Manager Managed Apple ID Auto-Provisioning (Government / GCC High)

This is the **Government (GCC High)** edition of the solution that integrates Microsoft Entra ID with Apple Business Manager (ABM) so that Managed Apple IDs and Apple device enrollment are driven from your directory. It is written for an administrator working in the **US sovereign cloud**, using the `*.us` admin portals and the US Gov Microsoft Graph endpoint. It follows the standard deployment model — **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** — with a gate at the end of each phase.

Read this first — the most important GCC High difference

Before anything else: **the Entra ID ■ Apple Business Manager _federation_ and _SCIM auto-provisioning_ link does not currently support national clouds (GCC High / DoD).** Apple's own documentation states, verbatim, "Integration with national clouds isn't currently supported," and the integration is built for the Commercial global sign-in service `login.microsoftonline.com`, not the sovereign `login.microsoftonline.us`. On the Microsoft side, the gallery provisioning app for ABM is not offered for provisioning in the US Government cloud.

What that means in plain terms:

- A **pure GCC High tenant cannot complete the "sign in to Apple with your Entra work account,

auto-created and kept in sync by SCIM" flow **the way a Commercial tenant can. This is a** feature parity gap**, not a portal-URL change.

- What **is** fully supported in GCC High is **Apple device management through Intune Government** —

linking ABM to Intune for **Automated Device Enrollment (ADE)** and **User Enrollment**, with Managed Apple IDs created by ABM's own tooling (manual, SFTP, or ABM API) rather than by the Entra federation/SCIM connector.

This solution therefore guides you to **implement the supported GCC High path** and **explicitly call out the federation/SCIM step as a Commercial-only capability** to evaluate separately with your security and compliance team. Full citations are in `gcc-high-parity-notes.md`; do not skip it.

What this solution delivers

- A clear **decision gate** up front: confirm whether your requirement can be met in GCC High today,

or whether it depends on the unsupported federation/SCIM link.

- The **supported GCC High deployment**: ABM linked to **Intune Government** (`endpoint.microsoft.us`)

for Apple ADE/User Enrollment, with Managed Apple ID lifecycle handled through ABM.

- **Five PowerShell scripts** that connect to the **US Gov** Graph endpoint

(`Connect-MgGraph -Environment USGov`) to discover readiness, validate configuration, report state, safely remediate, and build the pilot scope group.

- A **parity reference**, **testing gates**, and **troubleshooting** tuned to GCC-High-specific

behaviour.

Commercial vs GCC High — what's different

This section is the orientation every government administrator needs. The differences are not cosmetic — one of them changes what is achievable at all.

1. Feature parity — the federation/SCIM link (the big one)

	Commercial	GCC High
Sign in to Apple devices with Entra **work credentials** (federation)	Supported via `login.microsoftonline.com`	**Not supported** — Apple excludes national clouds
SCIM auto-provisioning of Managed Apple IDs from Entra	Supported (gallery provisioning app)	**Not supported / verify** — gallery app not offered for US Gov provisioning
ABM ■ **Intune** device enrollment (ADE, User Enrollment)	Supported	**Supported** via Intune Government
Managed Apple ID creation via **ABM** (manual / SFTP / ABM API)	Supported	Supported (ABM is a single global service, not cloud-bound)

Because the top two rows are the literal title of this solution ("Entra ID integration with ABM for account auto-provisioning"), a government reader must understand up front that the **Entra-driven** auto-provisioning is the gated capability, while the **device-management outcome** is achievable through the supported Intune path.

2. Admin portals — every URL changes to `*.us`

You do all supported configuration in the sovereign portals, never the Commercial ones:

- Entra admin center → **entra.microsoft.us**
- Intune / Endpoint admin → **endpoint.microsoft.us**
- Microsoft 365 admin → **portal.office365.us**
- Azure portal → **portal.azure.us**
- Sign-in authority → **login.microsoftonline.us**
- Microsoft Graph → **graph.microsoft.us**

Using a Commercial URL either fails to authenticate your GCC High credentials or silently targets the wrong tenant.

3. Tooling — Graph must be told it is US Gov

Every script and any manual Graph call must specify the sovereign environment. In the Microsoft Graph PowerShell SDK that is `Connect-MgGraph -Environment USGov` (which targets `graph.microsoft.us`). Omitting it authenticates against the Commercial endpoint and fails.

4. Licensing — Government SKUs

Licensing uses **Government** SKUs (for example Microsoft 365 **G3/G5 Gov** and **Intune for Government**) rather than the Commercial E3/E5 line. Confirm your tenant carries the Government licensing that includes Entra ID P1/P2 and Intune Government.

5. ABM itself is unchanged

Apple Business Manager has **no sovereign variant** — there is one global `business.apple.com`. The sovereign constraint lives entirely on the **Entra-side federation/SCIM link into ABM**. So Managed Apple IDs, MDM server tokens, and device sync tokens are configured in the same ABM you would use for Commercial; only the **identity link into it** is subject to the national-cloud limitation.

The deployment model (Assess → Configure → Pilot → Validate → Expand)

- **Assess** — inventory the tenant, licensing, roles, ABM account, and — critically — decide whether

your requirement can be met on the supported GCC High path or depends on the unsupported federation/SCIM link. Gate: no configuration until the decision is recorded.

- **Configure** — link ABM to **Intune Government**, establish the MDM push certificate and ADE/User

Enrollment tokens, and define the Managed Apple ID approach ABM will use. Gate: tokens valid, no errors in `endpoint.microsoft.us`.

- **Pilot** — enrol a small, named group of Apple devices/users (built by

`ABMPROVG-Set-ScopeTargeting.ps1`). Gate: pilot devices enrol and receive policy.

- **Validate** — run the validation and report scripts; confirm enrollment, Managed Apple ID sign-in

behaviour, and policy delivery against pass/fail criteria. Gate: all checks pass.

- **Expand** — widen the scope group in controlled waves. Gate: each wave green before the next.

Named objects you will create and reuse

To keep the deployment repeatable, create these once and reuse them by name throughout:

- `ABM-Apple-Pilot-Devices` — the pilot scope group (built by the scope-targeting script).
- `ABM-Apple-Enroll-Users` — the user group scoped for Apple enrollment.
- `ABM-ADE-Profile-Gov` — the Automated Device Enrollment profile in Intune Government.
- `ABM-Managed-AppleID-Standard` — your documented standard for how Managed Apple IDs are named

and created in ABM.

Who this is for and what "done" looks like

For a novice-friendly government administrator with the required roles. You are done when: the GCC-High-vs-Commercial decision is recorded; ABM is linked to Intune Government with valid tokens; a named pilot group of Apple devices enrols and receives policy; the validation script reports all checks green; and the federation/SCIM parity gap is documented for stakeholders so no one expects a capability the sovereign cloud does not offer today.

Sovereign-cloud capabilities change. Re-verify the cited Apple and Microsoft pages (see `gcc-high-parity-notes.md`) before implementation.

Requirements — Entra ID + ABM Managed Apple ID Auto-Provisioning (Government / GCC High)

The category-level prerequisites are in `global-requirements-entra-id-solutions.md` — confirm those first (Entra tenant, licensing, admin roles, break-glass accounts, network access). This document adds the requirements **specific to this solution on GCC High**, each with how to **verify** it and how to **implement it if missing**.

> *Everything below assumes the **US sovereign cloud**. Use the `*.us` portals and `Connect-MgGraph -Environment USGov`. Confirm the parity gap in `gcc-high-parity-notes.md` before you spend money or time — some requirements only matter if you pursue a path GCC High does not support today.*

R0. (Gate) Decision — supported path vs unsupported federation/SCIM link

Description: A recorded decision on whether your requirement can be met on the **supported GCC High path** (ABM ■ Intune Government enrollment) or depends on the **unsupported** Entra ■ ABM federation/SCIM link (national clouds excluded by Apple). Why: This decision determines whether the rest of the deployment is achievable in GCC High today. Skipping it risks configuring toward a capability that cannot complete. Verify: A short written statement citing `gcc-high-parity-notes.md`, approved by the project owner. Implement: Read `gcc-high-parity-notes.md`, re-check the Apple federated-authentication page live, confirm with your security/compliance team, and record the decision before any configuration.

R1. GCC High tenant with Government licensing

Description: An active GCC High Entra tenant carrying Government SKUs — Microsoft 365 **G3/G5 Gov** (or equivalent) that includes **Entra ID P1/P2** and **Intune for Government**. Why: Apple enrollment through Intune Government requires the Government Intune service and Entra P1/P2 for scoping and Conditional Access. Verify: Sign in to `https://portal.office365.us` → `Billing → Licenses`; confirm Government SKUs with Intune and Entra P1/P2. Sign in to `https://entra.microsoft.us` as an administrator. Implement: Assign Government licensing (M365 G3/G5 Gov or EMS Gov + Intune for Government).

R2. Administrative roles (sovereign)

Description: **Intune Administrator** (for the Apple enrollment configuration) and a **Global Administrator** available for one-time consent steps; least privilege thereafter. Why: Linking ABM to Intune and creating

enrollment profiles requires Intune Administrator; some one-time trust/consent actions require Global Administrator. Verify: `entra.microsoft.us` → `Roles and administrators`; confirm the operator holds Intune Administrator and that a Global Administrator is available. Implement: Assign **Intune Administrator** to the operator; reserve **Global Administrator** for one-time consent, then step down to least privilege.

R3. Break-glass accounts (sovereign)

Description: Two cloud-only emergency-access Global Administrator accounts (`@onmicrosoft.us` / your `.onmicrosoft.com` sovereign vanity), excluded from broad Conditional Access/MFA lockout. Why: Enrollment and Conditional Access changes can lock out admins; break-glass restores access. Verify: `entra.microsoft.us` → `Users`; confirm two cloud-only accounts with stored strong credentials and sign-in monitoring. Implement: Create two cloud-only Global Administrator accounts, exclude them from broad policies, and store credentials securely.

R4. Apple Business Manager account

Description: An active Apple Business Manager account at `business.apple.com` with an administrator who can create tokens and manage Managed Apple IDs. Why: ABM is where the MDM server (Intune) is linked, ADE tokens are issued, and Managed Apple IDs live. ABM is a single global Apple service — there is no sovereign variant. Verify: Sign in to `https://business.apple.com` as an ABM Administrator; confirm the organization is enrolled and verified. Implement: Enrol your organization in Apple Business Manager and complete Apple's verification. Note (**verify for GCC High**): ABM is not a sovereign service; the sovereign constraint is only on the Entra federation/SCIM link into ABM, not on ABM itself.

R5. Intune Government configured for Apple

Description: Intune Government reachable at `https://endpoint.microsoft.us`, with the Apple **MDM push certificate (APNs)** established. Why: Apple device enrollment through Intune requires a valid APNs certificate and the Intune Government service. Verify: `endpoint.microsoft.us` → `Devices` → `Enrollment` → `Apple` → confirm a valid, non-expired Apple MDM push certificate. Run `ABMPROVG-Discovery.ps1`. Implement: In `endpoint.microsoft.us`, create/download the APNs CSR, obtain the certificate from Apple, and upload it; record the Apple ID used so it can be renewed annually.

R6. ABM ■ Intune enrollment tokens (ADE / User Enrollment)

Description: An Automated Device Enrollment (ADE) token (and/or User Enrollment configuration) linking ABM to Intune Government. Why: This is the **supported** GCC High mechanism to enrol Apple devices and apply the Managed Apple ID experience through Intune. Verify: `endpoint.microsoft.us` → `Devices` → `Enrollment` → `Apple` → `Enrollment program tokens`; confirm a valid, non-expired token synced with ABM. Run `ABMPROVG-Validation.ps1`. Implement: In ABM, add Intune Government as an MDM server, download the public key, upload it to `endpoint.microsoft.us`, then upload the ABM token back to Intune; set the token to sync.

R7. Managed Apple ID approach (ABM-native, not Entra federation)

Description: A documented standard (`ABM-Managed-AppleID-Standard`) for how Managed Apple IDs are created — **manually, via ABM SFTP, or via the ABM API** — since Entra federation/SCIM is not available in GCC High. Why: With the federation/SCIM link unavailable, Managed Apple ID lifecycle must be handled by ABM's own tooling; a written standard keeps it repeatable and auditable. Verify: A documented naming and creation standard exists and matches how accounts are created in ABM. Implement: Choose an ABM-native creation method, document the naming convention and lifecycle (create/suspend/delete), and record it as `ABM-Managed-AppleID-Standard`.

R8. Network endpoints for US Government Intune + Apple

Description: Outbound HTTPS to the **US Government Intune** service endpoints and to Apple's ABM/APNs endpoints from managed devices and the admin workstation. Why: Enrollment, policy delivery, and ABM/APNs communication depend on these being reachable through sovereign filtering. Verify: From the admin workstation and a test device, reach `endpoint.microsoft.us`, `login.microsoftonline.us`, `graph.microsoft.us`, `business.apple.com`, and Apple's push endpoints; compare to Microsoft's published US Government Intune network endpoints. Implement: Allow the US Government Intune network endpoints and Apple ABM/APNs ranges through the proxy/firewall. Source: *Network endpoints for US government deployments (Intune)* — https://learn.microsoft.com/en-us/intune/intune-service/fundamentals/intune-us-government-endpoints

R9. Microsoft Graph PowerShell SDK targeting US Gov

Description: A workstation with the **Microsoft Graph PowerShell SDK** installed, able to run `Connect-MgGraph -Environment USGov` against `graph.microsoft.us`. Why: All five solution scripts connect to the sovereign Graph endpoint; the Commercial endpoint rejects GCC High credentials. Verify: `Get-Module Microsoft.Graph -ListAvailable` returns a version; `Connect-MgGraph -Environment USGov` succeeds. Run `ABMPROVG-Discovery.ps1`. Implement: `Install-Module Microsoft.Graph -Scope CurrentUser`; use `-Environment USGov` on every connect. Source: *Connect-MgGraph* — https://learn.microsoft.com/en-us/powershell/module/microsoft.graph.authentication/connect-mggraph

R10. Pilot scope groups

Description: The named groups `ABM-Apple-Pilot-Devices` and `ABM-Apple-Enroll-Users` for staged rollout. Why: Staged Pilot → Validate → Expand requires a small, named scope you can widen in waves. Verify: `entra.microsoft.us` → `Groups`; confirm both groups exist. Run `ABMPROVG-Set-ScopeTargeting.ps1 -WhatIf`. Implement: Create the groups (or let `ABMPROVG-Set-ScopeTargeting.ps1` create/populate the pilot group with `-WhatIf` removed after review).

The deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

Implementation Guide — Entra ID + ABM Managed Apple ID Auto-Provisioning (Government / GCC High)

Follow these steps in order, using the **US sovereign portals** (`*.us`) throughout. Each phase ends with a **gate** — do not proceed until it is met. This guide implements the **supported GCC High path** and flags the **unsupported Entra ■ ABM federation/SCIM** capability inline where a Commercial guide would tell you to configure it.

> *Parity reminder (read `gcc-high-parity-notes.md`): Apple states "Integration with national clouds isn't currently supported," so the Entra-driven federation + SCIM auto-provisioning of Managed Apple IDs **does not complete on a GCC High tenant today**. The device-management outcome is reached through **Intune Government**. Every "auto-provisioning" step below is therefore the **ABM-native + Intune** path, not the Commercial gallery/SCIM path.*

Phase 1 — Assess

Step 1.1 — Record the parity decision (Gate R0)

Before touching any portal, read `gcc-high-parity-notes.md`, re-check the Apple federated authentication page, and record — in writing, approved by the owner — whether your requirement is met by the supported GCC High path or depends on the unsupported federation/SCIM link. If it depends on federation/SCIM, **stop** and escalate the parity gap; do not attempt to force the Commercial flow against `login.microsoftonline.us`.

Step 1.2 — Confirm tenant, licensing, roles

1. Sign in to `https://portal.office365.us` → `Billing` → `Licenses`; confirm Government SKUs including

Intune Government and Entra ID P1/P2.

1. Sign in to `https://entra.microsoft.us` → `Roles and administrators`; confirm the operator holds

Intune Administrator and a **Global Administrator** is available for one-time consent.

1. Confirm two break-glass accounts exist (`entra.microsoft.us` → `Users`).

Step 1.3 — Run discovery

On the admin workstation, run the read-only discovery script:

```
powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Discovery.ps1
```

It connects with `Connect-MgGraph -Environment USGov` and reports tenant, licensing signals, Apple enrollment configuration presence, and the pilot groups. Resolve any red items before configuring.

Gate (Assess): parity decision recorded; licensing/roles/break-glass confirmed; discovery shows no blocking gaps.

Phase 2 — Configure

Step 2.1 — Establish the Apple MDM push certificate (APNs)

1. In `https://endpoint.microsoft.us` → `Devices` → `Enrollment` → `Apple`.
2. Under **Apple MDM Push certificate**, download the CSR, obtain the certificate from Apple's push

portal, and upload it.

1. Record the **Apple ID** used (you must renew with the same Apple ID annually). Name your record consistently so the renewal owner can find it.

Step 2.2 — Link Apple Business Manager to Intune Government (ADE token)

1. In `https://endpoint.microsoft.us` → `Devices → Enrollment → Apple → Enrollment program tokens →

Add`.

1. Download the Intune **public key**.

2. In `https://business.apple.com` → `Preferences → MDM Servers → Add`, name the server

`Intune-Gov-MDM`, and upload the Intune public key.

1. Download the ABM **server token** and upload it back into Intune to complete the link.

2. Set the token to **sync** on a schedule.

> *This is the step where a Commercial guide would instead (or additionally) add the **"Apple Business Manager" provisioning gallery app** in Enterprise Applications and configure **SCIM** with a tenant URL + secret token from ABM to auto-provision Managed Apple IDs. **In GCC High, that gallery provisioning app is not offered for the US Government cloud (verify live in `entra.microsoft.us`), and Apple does not support the federation that underpins it.** Do not attempt it here — use the ABM-native Managed Apple ID approach in Step 2.4.*

Step 2.3 — Create the Automated Device Enrollment profile

1. In `endpoint.microsoft.us` → `Devices → Enrollment → Apple → Enrollment program tokens` → select

your token → `Profiles → Create profile`.

1. Name it `ABM-ADE-Profile-Gov`, choose supervised/user-affinity settings per your policy, and assign it (initially unassigned or to a tiny test group; broaden in Pilot).

Step 2.4 — Define the Managed Apple ID approach (ABM-native)

Because Entra federation/SCIM is unavailable in GCC High, create Managed Apple IDs with ABM's own tooling per your documented `ABM-Managed-AppleID-Standard`:

- **Manual** — for small pilots, create Managed Apple IDs directly in `business.apple.com → Accounts`.
- **SFTP** — for bulk, use ABM's SFTP directory-sync feed (ABM-side, independent of Entra cloud).
- **ABM API** — for automation, use the Apple Business Manager API.

Record the naming convention and the create/suspend/delete lifecycle so it is auditable.

Gate (Configure): APNs certificate valid; ABM ■ Intune Government token valid and syncing; `ABM-ADE-Profile-Gov` created; Managed Apple ID standard documented. Run:

```
powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Validation.ps1
```

All configuration checks should pass before piloting.

Phase 3 — Pilot

Step 3.1 — Build the pilot scope

Review, then apply, the pilot scope group:

```
powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Set-ScopeTargeting.ps1 -WhatIf
```

Inspect the planned membership of `ABM-Apple-Pilot-Devices` / `ABM-Apple-Enroll-Users`, then re-run **without** `-WhatIf` to apply.

Step 3.2 — Assign the ADE profile to the pilot

Assign `ABM-ADE-Profile-Gov` to `ABM-Apple-Pilot-Devices` in `endpoint.microsoft.us`.

Step 3.3 — Enrol pilot devices

Wipe/prepare a small set of ABM-assigned Apple devices and run them through Setup Assistant. They should appear as ADE-enrolled in Intune Government and receive the assigned profile/policy. If using Managed Apple IDs, confirm sign-in with an ABM-created Managed Apple ID (not an Entra work account — that federation path is unavailable in GCC High).

Gate (Pilot): pilot devices enrol via ADE, appear in `endpoint.microsoft.us`, and receive policy.

Phase 4 — Validate

Step 4.1 — Run validation and report

```
powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Validation.ps1
powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Report.ps1
```

Confirm: token valid and syncing; APNs valid; pilot devices enrolled; profile applied; Managed Apple ID sign-in behaves per your standard. See `05-testing.md` for the pass/fail matrix.

Step 4.2 — Remediate safely if needed

If validation flags a fixable drift (for example a missing group assignment), preview and apply the safe remediation:

```
powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Remediation.ps1 -WhatIf
```

Review the planned actions, then re-run without `-WhatIf`.

Gate (Validate): every check in `05-testing.md` is green.

Phase 5 — Expand

Step 5.1 — Widen in waves

Grow `ABM-Apple-Pilot-Devices` (or graduate to a production group) in controlled waves using `ABMPROVG-Set-ScopeTargeting.ps1`. After each wave, re-run `ABMPROVG-Validation.ps1` and `ABMPROVG-Report.ps1` and only proceed when green.

Step 5.2 — Operationalize

• Calendar the **APNs certificate** and **ABM token** renewals (both expire annually).

• Keep the parity gap documented for stakeholders so no one expects Entra federation/SCIM auto-provisioning that GCC High does not offer today.

• Re-check the Apple federated-authentication page periodically — Apple's wording ("isn't *currently* supported") implies national-cloud support could be added later.

Gate (Expand): each wave green before the next; renewals scheduled; parity documented.

GCC-High-specific caveats (inline summary)

• **Federation/SCIM:** not supported for national clouds (Apple) — do not configure the Commercial gallery/SCIM flow; it will not complete.

• **Portals:** every URL is `*.us`; a Commercial URL fails or targets the wrong tenant.

• **Graph:** every script/manual call uses `-Environment USGov` (`graph.microsoft.us`).

• **Licensing:** Government SKUs (Intune Government, Entra P1/P2).

• **ABM:** single global service; no `.us` variant — configure it as you would for Commercial, but only the Intune link (not Entra federation) applies.

Re-verify the cited Apple and Microsoft pages (see `gcc-high-parity-notes.md`) before implementation; sovereign-cloud capabilities change.

Testing & Validation — Entra ID + ABM Managed Apple ID Auto-Provisioning (Government / GCC High)

Use this pass/fail matrix at each phase gate. Run the scripts with `Connect-MgGraph -Environment USGov` (the scripts do this for you). All portal checks use the `*.us` sovereign URLs.

> *Reminder: on GCC High the **Entra ■ ABM federation/SCIM auto-provisioning** path is **not supported** (Apple excludes national clouds). Tests below validate the **supported** path (ABM ■ Intune Government enrollment) — there is deliberately **no** test for SCIM provisioning success, because that connector is not available in the US Government cloud.*

T0. Parity decision recorded

• **Check:** A written, approved decision exists stating the requirement is met by the supported GCC High path (or the project is stopped/escalated if it depends on federation/SCIM).

• **Pass:** Decision document present and approved, citing `gcc-high-parity-notes.md`.

• **Fail:** No decision recorded — do not proceed.

T1. Sovereign sign-in and Graph environment

• **Check:** `Connect-MgGraph -Environment USGov` succeeds; the connected context shows a `graph.microsoft.us` / US Gov environment.

• **Pass:** Connected to the US Gov Graph endpoint with the expected tenant ID.

• **Fail:** Auth error, or connection lands on the Commercial endpoint (you omitted `-Environment USGov`).

T2. Licensing and roles

• **Check:** `portal.office365.us` shows Government SKUs with Intune Government and Entra P1/P2; the operator holds Intune Administrator.

• **Pass:** Correct Government licensing and roles present.

• **Fail:** Commercial SKUs, missing Intune Government, or insufficient role.

T3. Apple MDM push certificate (APNs)

• **Check:** `endpoint.microsoft.us` → `Devices` → `Enrollment` → `Apple` shows a valid, non-expired APNs certificate; `ABMPROVG-Validation.ps1` reports it present.

• **Pass:** Certificate valid, expiry > 30 days out, renewal Apple ID recorded.

• **Fail:** Missing/expired certificate, or unknown renewal Apple ID.

T4. ABM ■ Intune Government token

• **Check:** `endpoint.microsoft.us` → `Enrollment program tokens` shows a valid, non-expired token named for the ABM link (`Intune-Gov-MDM`), last sync recent; `ABMPROVG-Validation.ps1` confirms.

• **Pass:** Token valid, syncing, expiry > 30 days out.

• **Fail:** Token missing, expired, or not syncing.

T5. ADE profile exists and is assigned

• **Check:** `ABM-ADE-Profile-Gov` exists and is assigned to `ABM-Apple-Pilot-Devices`.

• **Pass:** Profile present and assigned to the pilot scope.

• **Fail:** Profile missing or unassigned.

T6. Pilot enrollment succeeds

• **Check:** A wiped, ABM-assigned pilot device runs Setup Assistant and appears ADE-enrolled in

`endpoint.microsoft.us`, receiving the assigned profile/policy.

- **Pass:** Device enrolls automatically and shows compliant/managed with policy applied.
- **Fail:** Device does not enroll automatically, or does not receive policy.

T7. Managed Apple ID sign-in (ABM-native)

- **Check:** Signing in with an **ABM-created Managed Apple ID** behaves per

`ABM-Managed-AppleID-Standard`.

- **Pass:** Managed Apple ID sign-in works as designed.
- **Fail (expected on the federation path):** Attempting to sign in with an **Entra work account via

federation **fails — this is** expected** in GCC High and is not a defect; it confirms the parity gap. Do not log this as a bug.

T8. Scope targeting is deterministic

- **Check:** `ABMPROVG-Set-ScopeTargeting.ps1 -WhatIf` lists the intended pilot membership with no

unexpected additions/removals.

- **Pass:** Planned membership matches intent; applying it yields exactly that.
- **Fail:** Unexpected members, or `-WhatIf` output differs from applied result.

T9. Reporting is accurate

- **Check:** `ABMPROVG-Report.ps1` output matches what the portals show (token, APNs, enrolled pilot

devices, group membership).

- **Pass:** Report matches portal reality.
- **Fail:** Report diverges from the portal — investigate before trusting it.

T10. Remediation is safe by default

- **Check:** `ABMPROVG-Remediation.ps1` runs in `-WhatIf` mode by default and only changes state when

`-WhatIf` is removed.

- **Pass:** Default run makes no changes and previews actions clearly.
- **Fail:** Any change occurs without explicitly removing `-WhatIf`.

Phase gate summary

Phase	Must pass
Assess	T0, T1, T2

Phase	Must pass
Configure	T3, T4, T5
Pilot	T6, T7, T8
Validate	T9, T10 (plus re-confirm T3–T6)
Expand	Re-run T4–T9 after each wave

Record the result (pass/fail + date) for each test at each gate so the deployment is auditable.

Troubleshooting — Entra ID + ABM Managed Apple ID Auto-Provisioning (Government / GCC High)

GCC-High-specific symptoms and fixes. Most issues fall into one of three buckets: you hit a Commercial endpoint, you expected the unsupported federation/SCIM path, or a token/certificate expired. Always work in the `*.us` portals and connect Graph with `-Environment USGov`.

"Sign in to Apple with the Entra work account fails" / federation won't turn on

- **Cause:** This is the **expected parity gap**, not a defect. Apple states "Integration with national

clouds isn't currently supported," and the federation is built for the Commercial global service `login.microsoftonline.com`, not `login.microsoftonline.us`.

- **Fix:** Do not attempt to force the Commercial federation flow against the sovereign endpoint. Use

ABM-native Managed Apple IDs (manual / SFTP / ABM API) and enrol devices via **Intune Government ADE/User Enrollment**. See `gcc-high-parity-notes.md`. If the requirement truly depends on federation, escalate the parity gap — it is not solvable in a pure GCC High tenant today.

"Apple Business Manager provisioning app isn't in my Enterprise Applications gallery"

- **Cause:** The ABM provisioning gallery app is **not offered for provisioning in the US Government

cloud**. Its absence in `entra.microsoft.us` is expected.

- **Fix:** Do not search for a workaround gallery app. Provision Managed Apple IDs with ABM's own

tooling per `ABM-Managed-AppleID-Standard`. (**Verify for GCC High** live — gallery availability changes over time.)

`Connect-MgGraph` fails or connects to the wrong tenant

- **Cause:** You omitted `-Environment USGov`, so the SDK authenticated against the Commercial endpoint

and rejected your GCC High credentials (looks like an auth failure), or silently used a Commercial context.

- **Fix:** Always run `Connect-MgGraph -Environment USGov`. Optionally set

`$PSDefaultParameterValues['Connect-MgGraph:Environment'] = 'USGov'` in your profile. Confirm the connected context targets `graph.microsoft.us`.

Portal shows nothing / "you don't have access"

- **Cause:** You opened a Commercial URL (`entra.microsoft.com`, `endpoint.microsoft.com`) instead of

the sovereign `*.us` equivalent, or lack the Intune Administrator role.

- **Fix:** Use `entra.microsoft.us`, `endpoint.microsoft.us`, `portal.office365.us`. Confirm the

operator holds **Intune Administrator** in `entra.microsoft.us → Roles and administrators`.

Devices don't auto-enrol during Setup Assistant

- **Cause:** The ABM ■ Intune token isn't syncing, the device isn't assigned to the Intune MDM server

in ABM, or `ABM-ADE-Profile-Gov` isn't assigned to the pilot group.

- **Fix:** In `endpoint.microsoft.us → Enrollment program tokens`, confirm the token is valid and

synced; in `business.apple.com`, confirm the device is assigned to `Intune-Gov-MDM`; confirm `ABM-ADE-Profile-Gov` is assigned to `ABM-Apple-Pilot-Devices`. Run `ABMPROVG-Validation.ps1`.

Enrolled devices get no policy

- **Cause:** The device enrolled but isn't in the scoped group, or the APNs certificate is invalid.
- **Fix:** Verify APNs validity in `endpoint.microsoft.us → Apple → MDM push certificate`; confirm

group membership with `ABMPROVG-Set-ScopeTargeting.ps1 -WhatIf`.

APNs certificate or ABM token "expired"

- **Cause:** Both expire annually; a lapse breaks Apple enrollment and push.
- **Fix:** Renew before expiry. Renew the APNs certificate with the **same Apple ID** (a new Apple ID

forces re-enrollment of all devices). Re-download and re-upload the ABM token. Calendar both renewals with 30-day lead time.

`ABMPROVG-Report.ps1` disagrees with the portal

- **Cause:** Stale sync, insufficient Graph scopes, or a Commercial-context connection.
- **Fix:** Re-run after a sync; ensure you consented to the read scopes the script requests; confirm

the connection is US Gov (`-Environment USGov`). Trust the portal until the report matches.

Remediation changed something unexpectedly

- **Cause:** `-WhatIf` was removed before the preview was reviewed.
- **Fix:** Always run `ABMPROVG-Remediation.ps1 -WhatIf` and `ABMPROVG-Set-ScopeTargeting.ps1 -WhatIf`

first, read the planned actions, then re-run without `-WhatIf`. If an unwanted change occurred, reverse it in the portal and re-validate.

Network filtering blocks enrollment

- **Cause:** US Government Intune endpoints or Apple ABM/APNs ranges aren't allowed through the proxy.
- **Fix:** Allow the published **US Government Intune network endpoints** and Apple ABM/APNs ranges.

Source: https://learn.microsoft.com/en-us/intune/intune-service/fundamentals/intune-us-government-endpoints

When to escalate

- The requirement genuinely needs **Entra federation / SCIM auto-provisioning** — that is a national cloud limitation on Apple's side, not a configuration you can fix. Escalate with the citation from `gcc-high-parity-notes.md` and evaluate alternatives (ABM-native provisioning) with stakeholders.
- Any consideration of a separate Commercial/GCC tenant purely to obtain federation — route to your ISSO first; it may breach your CMMC/ITAR/DFARS compliance boundary.

Re-verify the cited Apple and Microsoft pages before implementation; sovereign-cloud capabilities change.

Best Practices — Entra ID + ABM Managed Apple ID Auto-Provisioning (Government / GCC High)

Practices that keep this deployment correct, auditable, and low-risk on the US sovereign cloud.

Set expectations on the parity gap first

- Communicate up front, in writing, that **Entra ■ ABM federation and SCIM auto-provisioning are not supported for national clouds **(Apple's statement). Stakeholders who expect "sign in to Apple with your GCC High work account, auto-provisioned by SCIM" will otherwise treat a documented limitation as a project failure. Frame the delivered outcome as** ABM ■ Intune Government enrollment **with** ABM-native Managed Apple IDs**.
- Re-check the Apple federated-authentication page periodically — the wording "isn't *currently* supported" implies it may change; revisit before any future re-scope.

Always target the sovereign cloud

- Use only `*.us` portals (`entra.microsoft.us`, `endpoint.microsoft.us`, `portal.office365.us`).

Bookmark them so no one accidentally lands on a Commercial URL.

- Every Graph call — script or manual — uses `Connect-MgGraph -Environment USGov`. Consider setting

`$PSDefaultParameterValues['Connect-MgGraph:Environment'] = 'USGov'` in the admin profile so the Commercial endpoint is never hit by accident.

Stage everything; never big-bang

- Follow **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** with the gates in `05-testing.md`.
- Keep a small, named `ABM-Apple-Pilot-Devices` group and widen in waves, validating between each.
- Use `-WhatIf` first on `ABMPROVG-Set-ScopeTargeting.ps1` and `ABMPROVG-Remediation.ps1` — review the

preview before applying.

Protect sign-in with break-glass

- Maintain two cloud-only break-glass Global Administrators excluded from broad Conditional

Access/MFA lockout, with monitored sign-ins. Enrollment and CA changes can lock admins out.

Name and reuse objects

- Create named objects once and reuse them: `ABM-Apple-Pilot-Devices`, `ABM-Apple-Enroll-Users`,

`ABM-ADE-Profile-Gov`, `Intune-Gov-MDM` (the ABM MDM server), and the documented `ABM-Managed-AppleID-Standard`. Consistent names make scripts, reports, and troubleshooting predictable.

Manage the two annual expiries

- Both the **Apple MDM push certificate (APNs)** and the **ABM enrollment token** expire yearly.

Calendar both renewals with 30-day lead time. Renew the APNs certificate with the **same Apple ID** — a different Apple ID forces re-enrollment of every device.

- Record the renewal owner and the Apple ID used, so a departing admin does not orphan the

certificate.

Keep Managed Apple ID lifecycle auditable

• With federation/SCIM unavailable, document how Managed Apple IDs are created (manual / SFTP / ABM

API), named, suspended, and deleted. Tie suspension/deletion to your joiner-mover-leaver process even though it is a manual/ABM-side step rather than an Entra-driven one.

Least privilege

• Use **Intune Administrator** for day-to-day enrollment work; reserve **Global Administrator** for

one-time consent, then step down. Do not run routine operations as Global Administrator.

Validate before you trust reporting

• Cross-check `ABMPROVG-Report.ps1` against the portals during the first few runs. Once you trust it,

use it as your recurring health snapshot after each expansion wave and before each renewal.

Compliance-boundary caution on any "hybrid tenant" workaround

• A design that uses a separate Commercial/GCC tenant solely to obtain ABM federation (with GCC High

for everything else) is **not a documented Microsoft path** and may violate your CMMC/ITAR/DFARS boundary. Treat it as a security/compliance decision to evaluate with your ISSO — never adopt it silently to "make federation work."

Re-verify sources before implementation

• Sovereign-cloud capabilities move. Before each deployment, re-read the cited Apple and Microsoft

pages in `gcc-high-parity-notes.md` and confirm nothing has changed.

Deliverables Summary — Entra ID + ABM Managed Apple ID Auto-Provisioning (Government / GCC High)

Everything included in this Government (GCC High) edition and what each piece is for.

Documents (source)

File	Purpose
`01-solution-overview.md`	What the solution delivers on GCC High, and the **Commercial vs GCC High — what's different** section (federation/SCIM parity gap up front).
`02-requirements.md`	Every GCC-High-specific prerequisite with **verify** and **implement-if-missing**, including the R0 parity decision gate.

File	Purpose
`03-implementation.md`	Ordered Assess → Configure → Pilot → Validate → Expand steps using `*.us` portals, with GCC-High caveats inline.
`05-testing.md`	Pass/fail matrix mapped to phase gates (no SCIM-success test — that path is unsupported in US Gov).
`06-troubleshooting.md`	GCC-High-specific symptoms and fixes, including the "federation won't complete" expected behaviour.
`07-best-practices.md`	Operating practices: sovereign targeting, staging, expiries, least privilege, compliance-boundary caution.
`08-deliverables-summary.md`	This file.
`executive-summary.md`	One-page summary for decision-makers, leading with the parity gap.
`gcc-high-parity-notes.md`	Cited reference for every Commercial-vs-GCC-High difference; the factual backbone.
`global-requirements-entra-id-solutions.md`	Category-level Entra prerequisites (shared).

Scripts (scripts/)

All scripts connect to the **US Gov** Microsoft Graph endpoint with `Connect-MgGraph -Environment USGov`. Read-only scripts make no changes; change scripts default to `-WhatIf`.

Script	Type	Purpose
`ABMPROVG-Discovery.ps1`	Read-only	Inventory tenant, licensing signals, Apple enrollment configuration, and pilot groups; surface readiness gaps.
`ABMPROVG-Validation.ps1`	Read-only	Check APNs certificate, ABM ■ Intune token, ADE profile, and pilot assignment against pass/fail criteria.
`ABMPROVG-Report.ps1`	Read-only	Produce a health snapshot (token/APNs expiry, enrolled pilot devices, group membership) for records and reviews.
`ABMPROVG-Remediation.ps1`	Change (**-WhatIf default**)	Safely fix common drift (e.g. missing group assignment) after previewing the planned actions.

Script	Type	Purpose
`ABMPROVG-Set-ScopeTargeting.ps1`	Change (**-WhatIf default**)	Build/populate the `ABM-Apple-Pilot-Devices` scope group for staged rollout.

How they fit the deployment model

- **Assess:** `ABMPROVG-Discovery.ps1`
- **Configure:** `ABMPROVG-Validation.ps1`
- **Pilot:** `ABMPROVG-Set-ScopeTargeting.ps1`
- **Validate:** `ABMPROVG-Validation.ps1`, `ABMPROVG-Report.ps1`, `ABMPROVG-Remediation.ps1`
- **Expand:** re-run `ABMPROVG-Set-ScopeTargeting.ps1`, then `ABMPROVG-Validation.ps1` + `ABMPROVG-Report.ps1`

The one thing to remember

On GCC High, **Entra ■ ABM federation and SCIM auto-provisioning are not supported** (Apple excludes national clouds). This edition delivers the **supported** outcome — ABM ■ Intune Government enrollment with ABM-native Managed Apple IDs — and documents the parity gap so expectations are set correctly. See `gcc-high-parity-notes.md` for citations.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

ABMPROVG-Discovery.ps1

Purpose: Inventories a GCC High tenant's readiness to enrol Apple devices via Apple Business Manager linked to Intune Government. Reports sign-in context, Apple enrollment configuration presence (APNs push certificate, ABM/ADE enrollment tokens), and the pilot scope groups. It never changes anything. Replace the sample group names with your-group-name values if you renamed them (defaults: ABM-Apple-Pilot-Devices, ABM-Apple-Enroll-Users). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Discovery.ps1 (Graph scopes: DeviceManagementServiceConfig.Read.All, DeviceManagementConfiguration.Read.All, Group.Read.All, Organization.Read.All; -Environment USGov) EXPECTED RESULTS A section-by-section readout with PASS/WARN/INFO lines. PASS means the item is present and healthy; WARN flags a gap to resolve before configuring. Confirms the connection is US Gov. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Discovery.ps1 (Graph scopes: DeviceManagementServiceConfig.Read.All, DeviceManagementConfiguration.Read.All, Group.Read.All, Organization.Read.All; -Environment USGov) EXPECTED RESULTS A section-by-section readout with PASS/WARN/INFO lines. PASS means the item is present and healthy; WARN flags a gap to resolve before configuring. Confirms the connection is US Gov. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>

ABMPROVG-Remediation.ps1

Purpose: Fixes a small set of low-risk configuration gaps for the supported GCC High path: ensures the pilot scope group exists and (optionally) assigns the Automated Device Enrollment profile to it. Defaults to preview mode (-WhatIf) and makes no change until you re-run without -WhatIf. It will not attempt any unsupported federation/SCIM action. Replace sample names with your-group-name / your-profile-name if renamed (defaults: ABM-Apple-Pilot-Devices, ABM-ADE-Profile-Gov). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Remediation.ps1 (preview only) powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Remediation.ps1 -WhatIf:$false (apply) (Graph scopes: Group.ReadWrite.All, DeviceManagementServiceConfig.ReadWrite.All; -Environment USGov) EXPECTED RESULTS In preview (default) each planned action prints as "WHATIF: ...". With -WhatIf:$false the actions are applied and confirmed. Review the preview before applying. .NOTES White-label — the repeatable way. Change actions are gated behind -WhatIf (on by default) against Microsoft Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Remediation.ps1 (preview only) powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Remediation.ps1 -WhatIf:$false (apply) (Graph scopes: Group.ReadWrite.All, DeviceManagementServiceConfig.ReadWrite.All; -Environment USGov) EXPECTED RESULTS In preview (default) each planned action prints as "WHATIF: ...". With -WhatIf:$false the actions are applied and confirmed. Review the preview before applying. .NOTES White-label — the repeatable way. Change actions are gated behind -WhatIf (on by default) against Microsoft Graph US Gov endpoints. #>

ABMPROVG-Report.ps1

Purpose: Produces a point-in-time health snapshot for records and phase reviews: US Gov connection, APNs and ABM token expiries, ADE-enrolled Apple device counts, and pilot group membership. Optionally writes a CSV/JSON copy. It changes nothing. Set your-output-folder with -OutputPath if you want a saved copy (default: no file, console only). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Report.ps1 powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Report.ps1 -OutputPath .\abmprovg-report (Graph scopes: DeviceManagementServiceConfig.Read.All, DeviceManagementManagedDevices.Read.All, Group.Read.All; -Environment USGov) EXPECTED RESULTS A readable snapshot table in the console (and optional .csv/.json files). Use it after each expansion wave and before each token/certificate renewal. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Report.ps1 powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Report.ps1 -OutputPath .\abmprovg-report (Graph scopes: DeviceManagementServiceConfig.Read.All, DeviceManagementManagedDevices.Read.All, Group.Read.All; -Environment USGov) EXPECTED RESULTS A readable snapshot table in the console (and optional .csv/.json files). Use it after each expansion wave and before each token/certificate renewal. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>

ABMPROVG-Set-ScopeTargeting.ps1

Purpose: Creates the named pilot scope group if missing and adds the members you supply (device or user object IDs, or the members of a seed group) so you can pilot then expand in controlled waves. Defaults to preview mode (-WhatIf) and makes no change until re-run without -WhatIf. Replace sample values with your-group-name and your-object-ids (default group: ABM-Apple-Pilot-Devices). Provide members via -MemberObjectId or -SeedGroupName. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Set-ScopeTargeting.ps1 -MemberObjectId '00000000-0000-0000-0000-000000000001' powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Set-ScopeTargeting.ps1 -SeedGroupName 'ABM-Apple-Enroll-Users' -WhatIf:$false (Graph scopes: Group.ReadWrite.All; -Environment USGov) EXPECTED RESULTS In preview each planned add prints as "WHATIF: add the object id". With -WhatIf:$false members are added and the resulting count is shown. Widen the group in waves, validating between each. .NOTES White-label — the repeatable way. Change actions are gated behind -WhatIf (on by default) against Microsoft Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Set-ScopeTargeting.ps1 -MemberObjectId '00000000-0000-0000-0000-000000000001' powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Set-ScopeTargeting.ps1 -SeedGroupName 'ABM-Apple-Enroll-Users' -WhatIf:$false (Graph scopes: Group.ReadWrite.All; -Environment USGov) EXPECTED RESULTS In preview each planned add prints as "WHATIF: add the object id". With -WhatIf:$false members are added and the resulting count is shown. Widen the group in waves, validating between each. .NOTES White-label — the repeatable way. Change actions are gated behind -WhatIf (on by default) against Microsoft Graph US Gov endpoints. #>

ABMPROVG-Validation.ps1

Purpose: Validates the supported GCC High configuration against clear pass/fail criteria: US Gov connection, Apple MDM push certificate validity, ABM/ADE enrollment token validity and sync, an Automated Device Enrollment profile, and the pilot scope group assignment. It changes nothing. Replace sample names with your-group-name / your-profile-name values if renamed (defaults:

ABM-Apple-Pilot-Devices, ABM-ADE-Profile-Gov). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Validation.ps1 (Graph scopes: DeviceManagementServiceConfig.Read.All, DeviceManagementConfiguration.Read.All, Group.Read.All; -Environment USGov) EXPECTED RESULTS A checklist of PASS/FAIL lines and a final summary count. All PASS means the configuration is ready to pilot/expand. Any FAIL identifies the exact item to fix before proceeding. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\ABMPROVG-Validation.ps1 (Graph scopes: DeviceManagementServiceConfig.Read.All, DeviceManagementConfiguration.Read.All, Group.Read.All; -Environment USGov) EXPECTED RESULTS A checklist of PASS/FAIL lines and a final summary count. All PASS means the configuration is ready to pilot/expand. Any FAIL identifies the exact item to fix before proceeding. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>